

# **Lecture 02b – Sampling designs II**

ENVX2001 Applied Statistical Methods

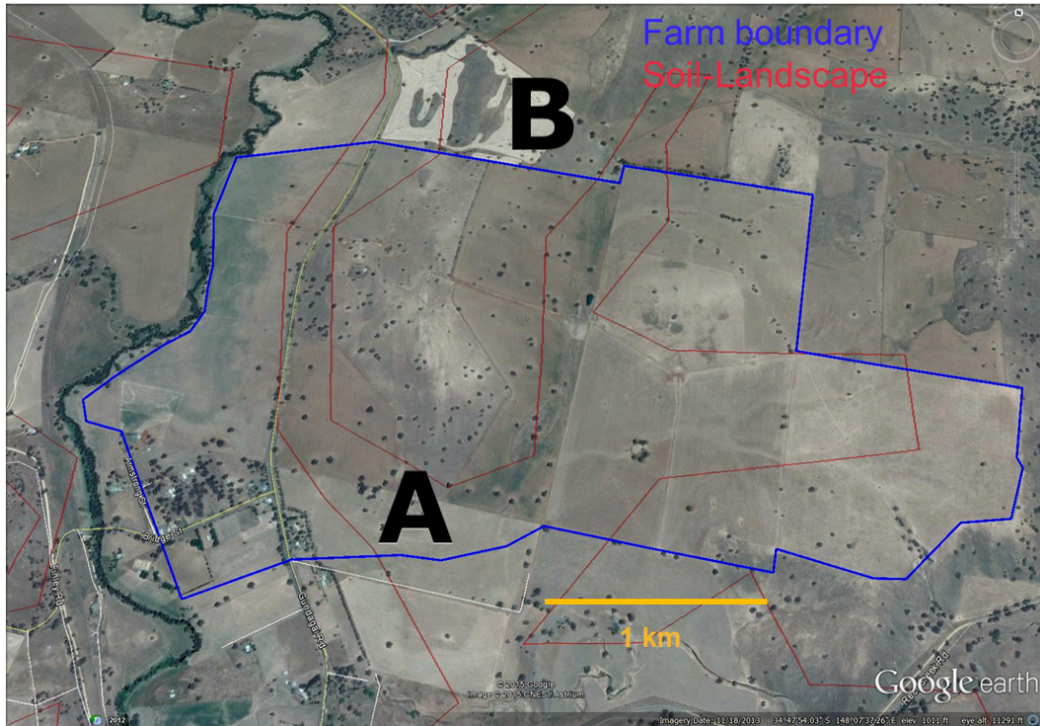
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Apr 2026

**Welcome back**

## Land types

The farmer decides to use stratified random sampling instead, dividing their property by land type.



- Land type A covers 62% of the area, land type B covers 38%
- They sample randomly within each land type

- For comparison, assume the same 7 locations happen to be selected

## Random sampling on a landscape

```
{ojs}  
//| echo: false  
import { samplingWidget } from "../../assets/js/sampling-widget.js"  
samplingWidget({  
  split: 0.62,  
  labelA: "Land type A",  
  labelB: "Land type B",  
  nStart: 7,  
  nMax: 30  
})
```

With two distinct land types, how do we guarantee both are represented in our sample?

# **Simple Stratified random sampling**

## Stratified random sampling

### 3 steps

1. **Divide** the population into **homogeneous** subgroups (strata).
2. **Sample** from each stratum using simple random sampling.
3. **Pool** (combine) the estimates from each stratum to get an overall population estimate.

### Example

If studying plant biodiversity in a national park:

- Divide the park into strata (e.g. forest, grassland, wetland)
- Take random samples within each habitat type
- Combine data to estimate overall biodiversity, weighting by each habitat's area

## Strata rules

### Strata are...

- **Mutually exclusive and collectively exhaustive**: every unit belongs to exactly one stratum, with no overlaps and nothing left out
- **Homogeneous**: units within a stratum should be more similar to each other than to the population as a whole
- **All sampled**: every stratum must be represented



## Advantages

Stratified random sampling addresses three problems at once:

- **Bias**: every stratum is sampled, so the sample is representative of the population
- **Accuracy**: each stratum is represented by a minimum number of units
- **Insight**: we can compare strata and make inferences about subgroups

## Does this make simple random sampling obsolete?

No. With large enough samples, the two methods converge. Simple random sampling is still a good default when strata are unknown or the population is fairly homogeneous.

## **Calculating stratified estimates**

## The workflow

Once we have our stratified sample, the calculations follow the same logic as simple random sampling, but each step must account for the stratified design.

1. **Pooled mean** ( $\bar{x}_s$ ): weight each stratum mean by the stratum's share of the population
2. **Pooled standard error**:  $SE(\bar{x}_s) = \sqrt{\sum w_i^2 \times \frac{s_i^2}{n_i}}$
3. **t-critical value**: based on  $df = n - L$  and  $\alpha = 0.05$
4. **Confidence interval**:  $\bar{x}_s \pm t_{n-L} \times SE(\bar{x}_s)$

## Accounting for strata using “weight”

- Each stratum contributes to the overall estimate in proportion to its size in the population.
- Most of the time, we use the stratum’s share of the total area (or population) as the weight.
- The overall population estimate is the sum of the weighted estimates from each stratum.

## Soil carbon: setting up the data

The farmer collects the same 7 measurements as before. This time, each value is assigned to its land type:

```
landA ← c(90, 78, 86, 71) # stratum A samples (62% of the area)
landB ← c(48, 56, 42)      # stratum B samples (38% of the area)
```

## Pooled mean $\bar{x}_s$

- The pooled mean is our best estimate of the overall population mean, taking into account the different stratum sizes.

$$\bar{x}_s = \sum_{i=1}^L \bar{x}_i \times w_i$$

We calculate the mean for each stratum ( $\bar{x}_i$ ), multiply by its weight ( $w_i$ ), and add them together.

## Calculating pooled mean: soil carbon example

We first define the weights  $w_i$  for each stratum based on their area:

```
weight ← c(0.62, 0.38) # 62% of area is land type A, 38% is land type B
```

Then we calculate the weighted mean:

```
weighted_mean ← mean(landA) * weight[1] + mean(landB) * weight[2]  
weighted_mean
```

```
[1] 68.86833
```

## Pooled standard error of the mean $SE(\bar{x}_s)$

$$SE(\bar{x}_s) = \sqrt{\sum_{i=1}^L w_i^2 \times \frac{s_i^2}{n_i}}$$

**i** What is different from simple random sampling?

- Instead of a single variance term, we sum the weighted variances from each stratum
- The  $w_i^2$  term accounts for the relative size of each stratum
- Each stratum contributes its own variance ( $s_i^2$ ) and sample size ( $n_i$ )



**It is not as complicated as it sounds...**

```
{ojs}  
//| echo: false  
import { stratifiedStepsWidget } from "../../assets/js/sampling-widget.js"  
stratifiedStepsWidget()
```

## *t*-critical value

## Degrees of freedom $df$

$$df = n - L$$

where  $n$  is the total number of samples and  $L$  is the number of strata.

- For stratified sampling, we lose one degree of freedom for each stratum mean we estimate
- **Example:** 12 samples across 3 strata gives  $df = 12 - 3 = 9$

## In R

```
df <- length(landA) + length(landB) - 2  
t_crit <- qt(0.975, df)  
t_crit
```

```
[1] 2.570582
```

## Putting it all together in R

```
varA ← var(landA) / length(landA) # variance of the mean for A
varB ← var(landB) / length(landB) # variance of the mean for B
weighted_var ← weight[1]^2 * varA + weight[2]^2 * varB
weighted_se ← sqrt(weighted_var)
ci ← c(
  L95 = weighted_mean - t_crit * weighted_se,
  u95 = weighted_mean + t_crit * weighted_se
)
ci
```

```
      L95      u95
61.04864 76.68803
```

**The payoff**

## Simple random vs. stratified random sampling

The farmer sampled the same 7 locations using a stratified design. How do the results compare to simple random sampling?

```
library(tidyverse)
# Manually printing the results below as SRS data is in previous lecture
compare <- tibble(
  Design = c("Simple Random", "Stratified Random"),
  Mean = c(67.29, 68.87),
  `Var (mean)` = c(50.80, 9.25),
  L95 = c(49.85, 61.05),
  U95 = c(84.73, 76.69),
  df = c(6, 5))
knitr::kable(compare)
```

| Design            | Mean  | Var (mean) | L95   | U95   | df |
|-------------------|-------|------------|-------|-------|----|
| Simple Random     | 67.29 | 50.80      | 49.85 | 84.73 | 6  |
| Stratified Random | 68.87 | 9.25       | 61.05 | 76.69 | 5  |

## Visual comparison of 95% confidence intervals

```
ggplot(compare, aes(x = Design, y = Mean)) +  
  geom_point(size = 3) +  
  geom_errorbar(aes(ymin = L95, ymax = U95), width = 0.2, linewidth = 1) +  
  labs(title = "95% Confidence Intervals by Sampling Design",  
        y = "Soil Carbon (tonnes/ha)",  
        x = "") +  
  theme_minimal(base_size = 14)
```

## 95% Confidence Intervals by Sampling Design



### Key differences

- Same locations, same measurements, different design
- Stratified sampling gives a variance about 5 times smaller

## Efficiency

### How much more precise is stratified sampling?

$$\text{Efficiency} = \frac{\text{Variance of SRS}}{\text{Variance of Stratified}}$$

- Efficiency > 1 means stratified sampling is more precise for the same sample size
- An efficiency of 5 means you would need 5 times as many SRS samples to match the stratified precision



## Efficiency (In R)

```
efficiency <- 50.80 / 9.25  
efficiency
```

```
[1] 5.491892
```

How many SRS samples would we need for the same precision?

```
round(7 * efficiency, 0)
```

```
[1] 38
```

About 38 samples with SRS to match what 7 stratified samples achieved.

## Tips on implementation

- The hardest part is choosing good strata (e.g. soil types, elevation bands, land use)
- Allocate samples in proportion to each stratum's size
- If one stratum is more variable, give it more samples

**A year later...**

## Repeating the survey

The farmer now has a solid baseline. They know their soil carbon is around 69 t/ha with a tight confidence interval. A year passes. They introduce cover cropping and want to know: **has soil carbon changed?**

This is the second type of observational study we met in the first half, a **monitoring study**. Instead of estimating a single value, we are estimating *change*.

To answer this question, the farmer measures the same property again.

## Same sites or new sites?

- Key decision: **how do we select sites for the second measurement?**
  1. Return to the **same sites?**
  2. Select completely **new sites?**
- This choice affects how we analyse the data

## Change in mean $\Delta\bar{x}$

- The difference between the means of the two sets of measurements.

$$\Delta\bar{x} = \bar{x}_2 - \bar{x}_1$$

where  $\bar{x}_2$  and  $\bar{x}_1$  are the means of the second and first set of measurements, respectively.

## Why same sites give better estimates

- Site 1 had 90 t/ha last year and 95 t/ha this year — a change of +5
- We do not care that site 1 is naturally carbon-rich, only that it went up
- Computing the difference at each site removes natural variation between sites
- This is why paired sampling gives tighter confidence intervals

## Paired vs independent sampling

- **Same sites** → paired  $t$ -test
- **Different sites** → two-sample  $t$ -test
- Paired sampling is usually more precise
- Each site is its own baseline, so natural differences drop out



## Analysing change in R

R handles both approaches through `t.test()`:

- Same sites: `t.test(after, before, paired = TRUE)`
- Different sites: `t.test(after, before, paired = FALSE)`

### R connection

`t.test()` with `paired = TRUE` performs a paired *t*-test. You will use this in Lab 02 to analyse monitoring data.

## The farmer's journey

Today we followed one farmer through four problems, and each problem needed a new statistical tool:

1. **“How do I choose where to sample?”** → Simple random sampling
2. **“How confident am I in the result?”** → Confidence intervals
3. **“Can I get a better estimate?”** → Stratified random sampling
4. **“Has anything changed?”** → Monitoring and paired comparisons

Each concept built on the last. The same logic (estimate, quantify uncertainty, improve) runs through every sampling study you will encounter in this course.

**Thanks!**

**Questions?**

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